

Undiscoverable meanings: Presupposition and the limits of word learning

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Abstract

Children are remarkable word learners, who exploit a variety of information sources — syntax, others’ attention and intention, general expectations about communication — to learn otherwise hidden meanings. Here we explore the learning of *presupposition triggers*, a ubiquitous but special class of words whose meanings cannot be learned from the usual information sources. In their syntactic distribution, they overlap with presupposition-free counterparts, and their meaning includes information that everyone in the conversation takes for granted, making it unattended and uninformative by definition. How, then, do children figure out that this information is part of a word’s meaning, let alone that it is presupposed?

We decompose the problem of presupposition learning into two components: *discovery* (registering that the presupposed content is part of the word’s meaning) and *packaging* (determining that it is presupposed rather than asserted). If all the child ever encountered were typical uses of triggers, the problem of discovery might indeed be unsolvable. We propose that presuppositions are instead discovered from atypical, informative uses. The packaging problem, in contrast, is solved using typical uses, with various sources of evidence bearing on it. Separating discovery from packaging makes the problem tractable and yields predictions about the course of learning, the evidence that drives it, and the intermediate meanings learners may entertain. Beyond development, the decomposition relocates the hard question, from how meanings get assigned their dimension to how presupposed content becomes representable at all. It offers an instructive test case for psychological accounts that tie word learning to communicative usefulness.

Keywords: presupposition; word learning; language acquisition; common ground

1 Introduction

Children learn words easily. They understand some words already in infancy (Bergelson and Swingley, 2012), produce their first word by their first birthday (Bloom, 2000; Waxman and Lidz, 2006), and command a considerable vocabulary by the preschool years (Snedeker, 2009). This is a remarkable accomplishment, because even when a learner tracks a word across situations, the situations themselves radically underdetermine the word’s meaning (Gleitman and Trueswell, 2020; Yu and Smith, 2007; Smith and Yu, 2008). Current consensus is that children’s success owes itself to other information sources they consult to restrict the hypothesis space. Learners track words’ syntactic profile and make assumptions about corresponding semantic correlates (Landau and Gleitman, 1985; Naigles, 1996; Gleitman et al., 2005; Fisher et al., 2010). They reason about speakers’ referential intentions and attention (Akhtar and Tomasello, 1996; Bloom, 2000; Frank and Goodman, 2012), and about how rational communication works in general (Grice, 1975; Clark, 1987; Diesendruck and Markson, 2001; Barner and Bachrach, 2010; Rasin and Aravind, 2021; Hacquard, 2023).

All of these different ways of learning words work for a common reason: each is a channel on which the target word meaning is systematically connected to something detectable — the word’s distribution, what the speaker attends to, the likely news the utterance contributes. A meaning is learnable, with this toolkit, only to the extent that it surfaces on at least one of these channels.

This paper is about a class of words for which every one of these channels fails, and fails for principled reasons, not because the child fails to recruit the information channels effectively. The problem is that these words encode, alongside their ordinary content, a meaning component p with three properties:

- (1) A word with a meaning component p
 - a. overlaps in syntactic distribution with counterparts that lack p ;
 - b. in typical uses, p is not what the speaker attends to or intends to convey;
 - c. in typical uses, p is information the audience already has.

Consider the learner who encounters such a word. Why should they assume the existence of p when it seems to leave no detectable trace? Why would they ever prefer a hypothesis with p when they always have available a simpler rival hypothesis, the same meaning without p , which fits every observation? In this paper, we articulate this learning challenge and explain why it is so hard. We then attempt to answer how a learner might in fact solve this problem and converge on meanings that encode p .

This is neither a hypothetical question nor a marginal phenomenon in the lexicon. Natural languages are full of words fitting the specification in (1), called presupposition triggers. When these words are used, part of their content is, by design, backgrounded and unattended (b) and already old information to all conversational participants (c). Their syntactic distribution also overlaps, in large part, with presupposition-free counterparts (a). Yet children learn these meanings; we don’t know how.

1.1 Background on presuppositions

Suppose we are talking about our friend Dana, and one of us utters (2).

- (2) Dana stopped playing chess.
- a. Dana does not play chess now. *asserted content*
 - b. Dana played chess in the past. *presupposition*

From this utterance you would gather both that Dana does not play chess now and that she used to. But the two pieces of information seem to play different roles. The first is the point of the utterance, the news. The second is background information the speaker assumes is already taken for granted among the conversational participants. The meaning of (2) divides into a foreground, its *asserted content*, and a background, its *presupposition*.

This difference tracks differences in compositional behavior — how these meanings interact with their linguistic surroundings. Presuppositions *project* (Karttunen, 1974; Heim, 1983). Negate the sentence, question it, or place it in the antecedent of a conditional, as in (3), and the presupposition that Dana used to play chess is still there, even if the original asserted content is not preserved.

- (3) a. It's not the case that Dana stopped playing chess.
b. Did Dana stop playing chess?
c. If Dana stopped playing chess, her club will be disappointed.
- Dana played chess in the past
↗ Dana does not play chess now

Pragmatically, the difference can be observed in the forms of protest they allow. A listener who responds to (2) with “No, she didn’t!” can only target the asserted content, denying that Dana quit, not the presupposition that she played before. To protest the presupposition, one must go out of one’s way, with something like the “Hey, wait a minute” move in (4) (Shanon, 1976; von Stechow, 2008).¹ This form of protest is odd when aimed at asserted content: of course you didn’t already know that Dana stopped; informing you of that was the point of the utterance.

- (4) Hey, wait a minute! I didn’t know Dana ever played chess!
#Hey, wait a minute! I didn’t know Dana doesn’t play chess now!

Presupposition triggers are ubiquitous across languages: definite descriptions (*the chess-player*), change-of-state verbs (*stop, continue*), additive particles (*too, again*), quantificational determiners (*both, neither*), factive predicates (*regret, forget, be happy that*), and many more (Levinson and Annamalai, 1992; Simons, 2001; Abrusán, 2011, 2016; Tonhauser et al., 2013). On the analysis of these words we adopt here for concreteness, lexical meanings are two-dimensional: a word can encode both an asserted

¹Here and throughout, # marks deviance. The symbol * marks ungrammaticality.

and a presupposed component (Heim and Kratzer, 1998). Asserted content must be true for the sentence to be true: for (2), that Dana is a current non-player. Presupposed content must be true for the sentence to be truth-evaluable at all: if Dana never played chess, uttering (2) is infelicitous. The two dimensions also differ in conversational function (Stalnaker, 1974, 2002). Assertions update the interlocutors’ *common ground*, the body of information mutually taken for granted, by adding to it. A cooperative speaker therefore asserts what the listener does not yet know. Presuppositions are *preconditions* on the update: they must already be common ground when the word is used. The two use conditions are thus mirror images. One must assert only what is new, and presuppose only what is known.

In practice, however, speakers sometimes violate the condition that presuppositions are known to the listener. Consider a speaker who says “I had to take my spouse to the doctor” to someone who did not previously know the speaker’s marital status. The presupposed content — that the speaker has a spouse — is now *informative* to the listener. Still, there is no ensuing conversation breakdown. Such cases succeed because listeners can *accommodate* the presupposition: they can tacitly adjust their beliefs by incorporating the presupposition, and proceed as if it had been common ground. The speaker, anticipating the listener’s cooperative stance, might then rely on their ability to accommodate and use sentences with unshared presuppositions to some strategic end. These informative uses of presuppositions will matter at several points below; we return to them in Sections 2.1 and 3.

1.2 The learning problem: discovery and packaging

Presuppositions are tied to individual words. Whatever else is true, the child must learn, for each word of her language, not only its asserted content but also whether it carries a presupposition and, if so, what that presupposition is. The obstacle is that presuppositions are, in their ordinary uses, uninformative, non-salient, and not what the speaker is attending to. These properties, which essentially define the category, make it so that presupposed content will not get picked up by the standard information channels for word learning.

Consider the quantifier *both*, which we will use as a running example. Assume the adult meaning is that of a universal quantifier carrying a domain cardinality presupposition (Ladusaw, 1982; Landman, 1989; Roberts, 1987), as in (5a). A learner encountering the word must locate this meaning in a vast hypothesis space, a relevant sample of which we enumerate in (5).

(5) Candidate meanings for *both*:

- a. $\llbracket \textit{both} \rrbracket(A)(B)$: presupposes $|A| = 2$; then true iff $A \subseteq B$ (target)
- b. $\llbracket \textit{both} \rrbracket(A)(B)$: true iff $A \subseteq B$ and $|A| = 2$ (conjunctive)
- c. $\llbracket \textit{both} \rrbracket(A)(B)$: presupposes $A \subseteq B$; then true iff $|A| = 2$ (inversely packaged)
- d. $\llbracket \textit{both} \rrbracket(A)(B)$: true iff $A \subseteq B$ (bare, $\approx$ all)
- e. $\llbracket \textit{both} \rrbracket(A)(B)$: true iff $|A \cap B| = 2$ (bare, $\approx$ two)
- f. $\llbracket \textit{both} \rrbracket(A)(B)$: true iff... (and so forth)

As these hypotheses reveal, converging on the target meaning requires the learner to succeed in solving two in principle separable problems. The first is realizing that the presupposition is part of the meaning in the first place. Call this the **discovery problem**. It is solved when the candidate hypotheses have both components encoded, and bare meanings are excluded. The second is the **packaging problem**, which is solved when each of the (already discovered) component meanings is mapped to the right dimension.

(6) The two problems of presupposition learning

- a. **The discovery problem:** for a trigger W whose target meaning comprises an asserted component a and a presupposed component p , the learner has to register that both components are part of W 's meaning. That is, they have to eliminate candidate meanings that encode only one component, such as (5d–e), in favor of candidates that encode both, such as (5a–c).
- b. **The packaging problem:** given a meaning for W that encodes both a and p , the problem of assigning each component to its dimension, that is, of eliminating the conjunctive meanings (5b) and inversely packaged meanings (5c) in favor of the target (5a), on which a is asserted and p presupposed.

Each problem raises two questions:

(7) Questions:

- a. *Channels:* What information in the child's experience could, in principle, provide evidence to solve the problem?
- b. *Learning:* Which of the available information sources do learners actually use?

To see the challenge in discovery, consider a prototypical input for learning *both*: a parent says “I put away both your toys” in a context where it is common knowledge that there are exactly two toys. In this situation, universal quantification is the news, and two-ness is mutually taken for granted. This use fits equally well with the target meaning (5a) and with a bare meaning (5d). They assign the utterance the same informational value, and simplicity would favor the bare meaning. We (and arguably the learner) can see nothing in the scene, the syntax, or the conversational dynamics that would favor the richer hypothesis.

In this way, the discovery problem is a worst-case variant of a more familiar one, called the subset problem in the acquisition of logical vocabulary (Gualmini and Schwarz, 2009; Piantadosi et al., 2016; Rasin and Aravind, 2021). When one candidate meaning is logically stronger than another, situations verifying the stronger meaning necessarily verify the weaker one, and naive positive-evidence learning cannot converge on the stronger meaning. Rasin and Aravind (2021) argue that reasoning about information strength is critical for solving the subset problem for expressions like *every*: there are attested uses that would have been redundant or trivial on the weaker construal, and a learner with Gricean expectations can exploit them. The problem with *both*, however, is that, although *both* is “stronger” than

all, that extra strength is packaged as a presupposition, that is, as information that is already known to the listener whenever the word is felicitously used. This means in most contexts of use, *both* and *all* are equally informative. The pragmatic escape route from the subset problem is unavailable here. Other general-purpose heuristics, like a preference for simplicity, might be actively harmful, as they favor bare meanings.

The packaging problem is of a different nature. Suppose the learner has registered the cardinality condition, so that the bare meaning is off the table. In typical contexts, the target and conjunctive meanings convey the same amount of information, but are nevertheless distinguishable. The two hypotheses differ in behavior under embedding and in use-conditional status. Only on the target meaning does the cardinality condition project through negation and questions, resist direct denial, and remain felicitous when redundant. Evidence of that kind, as we will show, is present.

In what follows, we first ground these two learning problems in empirical work. Then we turn to potential solutions. We first take up discovery, and eliminate, one by one, the routes available to a learner who only ever encounters typical uses. We conclude that from typical uses alone, presuppositional meanings might indeed be undiscoverable. Lacking a better solution, we drop the assumption that presuppositions are discovered from typical uses. Discovery instead requires *atypical* uses, those instances where a speaker uses a presuppositional word whose presupposition is not already taken for granted and is therefore informative. Once the relevant content is discovered, the packaging problem is more easily solved, via typical uses, with several channels of converging information.

2 Learners and the two problems

So far, we have provided a conceptual description of the learning problem and abstracted away from actual learners. What do we know about how presuppositions are learned? We do not aim here for comprehensiveness, both because triggers are heterogeneous, and because their acquisition has so far been studied piecemeal (for a recent review see Giunta and Pouscoulous, 2026). Instead, we focus on three case studies to ground our characterization. First, even preschoolers are highly sensitive to the presupposition/assertion distinction. This means they indeed face the challenge of learning presuppositions. Second, children often posit bare meanings as candidate hypotheses for presuppositional words. This shows that discovery is part of the problem they have to solve. Third, even adult learners who succeed with discovery posit conjunctive assertive meanings. This shows that packaging is a problem separable from discovery.

2.1 Children distinguish presupposed and asserted content

First, preschoolers are sensitive to the distinct use conditions governing assertion and presupposition. Aravind et al. (2023) tested whether preschoolers can identify the intended addressee of an utterance based on what potential listeners already knew and on whether the utterance presupposed or asserted the relevant information. For example, if a speaker says, “The bird that I got flew away!”, and one lis-

tener knows that the speaker got a bird while the other does not, an adult-like participant should select the former: presupposed information should be common ground between speaker and listener. If the speaker says instead, “I got a bird”, that same listener would be a poor choice: assertions should not be redundant. Both adults and preschoolers behaved in line with these expectations, preferring the listener who already knew the presupposition of the uttered sentence, and the listener who did *not* yet know its asserted content. If anything, children were stricter than adults. In a further experiment probing accommodation, four- and five-year-olds struggled when the context called for accommodating an informative presupposition. Children seemed to expect presuppositions to *always* be old information, with tolerance for informative presuppositions emerging only around six (Aravind et al., 2023). Well before school age, then, children distinguish the two dimensions of meaning: they expect presuppositions to be old news and assertions to be informative. Still, as we will see next, both discovery and packaging prove difficult for learners.

2.2 Discovering both components of *both*

Both and its counterpart *either* are presuppositional quantifiers that require their domains to have exactly two members. Modulo this presupposition, *both* is a universal quantifier (roughly *all*; see (5a)) and *either* an existential one (roughly *any*). (8) illustrates.

- (8) a. You can sign both of your kids up for the lesson.
presupposes: you have exactly two kids; *asserts*: you can sign them all up.
- b. You can sign either of your kids up for the lesson.
presupposes: you have exactly two kids; *asserts*: you can sign one up.

This quantifier pair is a useful testing ground because its meaning components are independently lexicalized in English. Duality is carried by the numeral *two*, universal force by *all*, and existential force by *any/some*. One can thus disentangle difficulties in presupposition learning from independent difficulties with the component pieces.

Torma et al. (2021) explored the development of these quantifiers and found that while children were essentially adult-like with *two*, *all*, and *any*, they were revealingly non-adult-like with *both* and *either*. Children treated *both* on par with *two* (i.e., in line with (5e)), suggesting that the cardinality component was registered but encoded as asserted. With *either*, children simply did not notice the presupposition at all: *either* was treated as no different from the non-presuppositional *any* in choice of domain. Thus, with these quantifiers, children seem to go through a stage where they entertain simplex, bare, purely assertive meanings comprising just one of their meaning components.

2.3 After discovery, packaging remains

The final piece of empirical evidence concerns adult learners, and it brings two insights. First, the discovery problem is hard to solve for adults, whose command of the presupposition/assertion distinction

can be taken for granted. Second, it shows that even after the discovery problem is solved, packaging can remain unsolved.

Alstott and Aravind (2026), following up on earlier work by Bade et al. (2024), trained adults on novel change-of-state verbs in a set of artificial learning experiments. Participants got evidence for a novel verb *wug*, associated with a complex change-of-state meaning: ‘[move up]_a [from the red line]_p’ in the first experiment and ‘[darken]_a [from gray]_p’ in the second experiment. In both cases, the initial state was meant to be presupposed and the change asserted.² Discovery conditions were ideal: the candidate content was perceptually present, and perfectly correlated with the word on every trial. Still, most participants never discovered the initial state, treating the verb instead as meaning simply ‘move up’ or ‘darken’. A small minority encoded the relevant meanings, but did so conjunctively, with both the initial state and change of state being packaged as ordinary asserted content. Thus, packaging can fail independently of discovery.

3 Routes to presuppositional meanings

Yet, learners do eventually discover presuppositions. In this section, we survey candidate routes by which this task may be accomplished. An important note before we begin: for no presupposition trigger do we currently have direct evidence about the mechanism by which children learn it. At best, what we have is evidence bearing on the weaker question (7a): where in the child’s experience there is correlational information that a suitably equipped learner could in principle exploit. This question — what in the input could reveal a trigger’s presupposition — has been pursued most thoroughly for factive *know* (Dudley, 2017; Dudley et al., 2017).

3.1 Routes to discovery

3.1.1 The direct route: tracking what is taken for granted

Proposal. Could a learner simply encode that some information is being taken for granted when a word is used, and track such co-occurrences across contexts? If there is such a statistical learning mechanism, we believe it could help solve discovery.³ Such a ‘direct’ route would need a statistical learner to establish a complicated set of contingencies with several parts. It has to find (1) that at uses of word *W*, the content *p* is common ground; (2) that *p*’s being common ground at those uses exceeds its base rate, since real presuppositions are common ground in other conversations without *W*; and (3) that no comparable regularity links *p* to other expressions.

²With change-of-state verbs, this mapping between component meanings and packaging seems to be stable: initial states tend to be presupposed, and changes asserted.

³Note that this route could help with packaging as well. The content would be discovered *because* it is established already and as such it could arrive already marked as presuppositional.

Assessment. Executing the strategy requires two things, and both requirements are, we think, prohibitive. The first concerns what needs to be tracked. It is not enough for the learner to note what is established at each use of each candidate word. To find what is established at uses of *both* above base rate, and more than at uses of arbitrary other words, the learner needs to track (a) how often each piece of content is common ground at uses of *both*, (b) at uses of other words, and (c) in conversation at large. The record to be maintained is thus not one entry per use of a word, but a full record of the common grounds of every conversation. It must also be kept at multiple levels of abstraction, since what recurs across uses of *both* is not any one proposition (e.g., there are 2 apples in the domain) but the two-ness of whatever domain is being quantified over (there are two NP-individuals in the domain, for whatever NP is the restrictor of *both*). Making matters worse, the body of established information is not a small, enumerable list. The common ground is open-ended, implicit, and comprising, in addition to conversation-specific details (where the interlocutors are, what day it is, their relationship), a host of general world knowledge (who the President is, that people have two legs, that there is exactly one sun). Keeping rates over all of this, always, is arguably harder than the word learning problem it was meant to help solve, and nothing we know about memory for conversational context suggests that children (or adults) keep records of this kind.

The second problem, of ‘spurious’ correlates, arises even if we grant the learner the full record. The base-rate comparison can eliminate standing facts from candidacy. That there is one sun and that people have two legs are common ground at every use of every word, and are screened out as non-diagnostic. And any particular accidental fact will, given enough data, be filtered out as well. But there might never be enough data to filter all of them at once. The sample of uses of *both* is finite, while the pool of candidate propositions is open-ended, and each common ground will inevitably be different from the last. So at any sample size, some propositions are elevated by accident — it happened to be morning at the observed uses, the dog happened to be in the room — and within the available sample, these spurious correlates pass the same tests as the two-ness of the domain. The two problems compound. The more of the established facts the learner considers, the more accidental correlates her sample will contain. The cues that can help prune spurious correlations, like salience, speaker attention, informativity, are precisely those that presupposition lacks. The true correlate is, in typical uses, no more salient, no more attended, and no more informative than the accidental ones.

Worse, correlates that are not part of the word’s meaning, but are related to it in a non-accidental way, will never be filtered out, no matter how much data accrues. Whatever circumstance tends to make the presupposition common ground inherits its statistical profile: it too co-occurs with the word stably, above base rate, and more than with arbitrary other words. Take *both* once more. *Both* is often used in relation to eyes, hands, etc., because duality of these body parts is always common ground. So, co-occurrence of *both* with body-related content will be stable, well above base rate, and above those of arbitrary other words. Yet *both* presupposes nothing about bodies. We see no obvious statistical way out. Presupposition is a condition on felicity, rather than a regularity of co-occurrence. It is content whose absence from the common ground makes a word’s use *defective* and not content whose absence makes

it less *likely*. It is unclear how statistics over occurrences, without statistics of marked violations, would ever draw that distinction.

Thus, as a general mechanism, direct tracking is a non-starter. But there is a restricted version of direct tracking that could help with a special subclass of triggers (Dudley, 2017). The presuppositions of some triggers have an anaphoric component: *too* does not merely require that its presupposition be common ground; it requires the presupposition to have been explicitly established in the immediately preceding discourse (Kripke, 2009). For such triggers, a strategy like the direct route becomes more available, with a crucial difference in what must be searched. The learner does not have to look at the open-ended body of established information. They could rely only on the conversational record, which is the short, ordered history of what was said and done in the context. This is a bounded record which has to be independently maintained, since the child needs it to participate in conversation anyway. Just looking at the conversational record, the diagnosticity problem becomes tractable. If every use of *too* is systematically preceded by certain linguistic material in the record, that regularity is detectable in a way that a regularity defined over everything taken for granted is not. It may be no accident that triggers like *too* are among the earliest acquired, often by age 2 (Hüttner et al., 2004; Höhle et al., 2009; Müller et al., 2011; Berger and Pouscoulous, 2014), though the connection remains circumstantial.

3.1.2 Learning by exclusion: presupposition from lexical contrast

Proposal. A second strategy stays within typical uses but shifts the evidential basis from the context to the lexicon. Learners expect distinct linguistic forms to have distinct meanings (Clark, 1987), and this expectation can be recruited for discovery. Suppose the child first masters the non-presuppositional member of a pair, for instance *all*. Now the child confronts *both*, a word with highly similar meaning and distribution. Why would the language have two words for the same thing? Noticing this could lead a learner to search for a hidden difference. In cases where a difference cannot be found in the asserted content, the child might conclude that it might be in the presupposition. Note that the route provides evidence only that a presupposition exists. It alerts the learner that there is something to find, without necessarily identifying it.

Assessment. The route is economical as it only requires an expectation of contrast that is independently attested in word learning. The route also makes a directional prediction that is consistent with the available evidence: the presuppositional member of a pair should be acquired only after its non-presuppositional counterpart. This is plausibly true, with *all* being learned before *both*, *any* before *either*, and *think* before *know* (Dudley et al., 2015; Torma et al., 2021).

But its starting conditions are rarely met. Genuinely minimal pairs are scarce. Even the best-case candidates — *both/all*, *know/think* — are not similar in all respects but one. For example, *all*, unlike *both*, combines with collective predicates (“All of the students met in the hall” vs. “#Both of the students met in the hall”). Likewise, *know* differs from *think* also in its ability to embed a question (“I know who won” vs. “*I think who won”). Crucially, such distributional differences, where they exist, need

not correlate with presuppositionality. A learner in search of a difference could reasonably localize the contrast at these places and stop, never positing a presupposition at all. Even if near-synonyms exist, it is not obvious that children always register them as such. The children in Torma et al. (2021) equated *both* with *two* and not with *all*, failing to identify the very similarity that is supposed to launch the inference.

3.1.3 Learning via competition: Maximize Presupposition

Proposal. Non-synonymy reasoning can be sharpened into a principled inference. There are circumstances in which speakers not only *can* use a presupposition trigger, but *must*. The principle, *Maximize Presupposition!* (Heim, 1991; Percus, 2006; Sauerland, 2008), says that when two forms differ in their presuppositions but would otherwise contribute the same new information to the common ground, speakers should choose the presuppositionally stronger one. This principle regulates presupposed content much as the Maxim of Quantity regulates asserted content, and its footprint in adult language is the oddness of presuppositionally weak choices where strong ones were available:

- (9) a. #A sun is shining.
- b. #Dana broke all her legs.

The sentences in (9) suggest, oddly, that the speaker does not share our assumptions that there is exactly one sun and that people have exactly two legs. Failure to use *the* and *both* signals that their presuppositions are not taken for granted.

A child equipped with this principle could use it in reverse. Suppose she notices that *both* and *all* are used for similar jobs in similar contexts. Suppose she further notices that *both* is not detectably stronger; across its uses, a *both*-sentence never conveys more news than the corresponding *all*-sentence would have. Why, then, does the language maintain the second form, and why do speakers ever choose it? If the child expects speakers to maximize presuppositions, the choice of *both* over *all* indicates that *both* carries a presupposition satisfied in the context, and she can eliminate the bare hypothesis. This expectation would convert the absence of new information, which is at the heart of the problem, into a signal. However, as with the previous route, the inference tells only *that* a word presupposes something satisfied in the context, not *what*.

Assessment. The route has substantial prerequisites. The child must have the principle in place, command the meaning of the competitor (*all*), and represent *both* and *all* as alternatives. There is some evidence for sensitivity to the principle in childhood. In a study by Aravind (2018), preschoolers showed a preference for using *another* over *a* when *another*'s additive presupposition was met. The alternative-set assumption, however, is not innocent. Whether an expression has a presuppositional competitor is itself a cross-linguistically arbitrary fact, as Chemla (2007) observes for French, which lacks a lexical counterpart of *both*.

This route improves on exclusion in one respect: it gives the learner a principled reason to posit a presupposition, not just a suspicion of hidden difference. But it inherits the diagnosticity problem of

Section 3.1.1: everything that is in the common ground is a possible candidate for that presupposition, with nothing to single out which background assumption is the one actually tied to the word.

3.1.4 Informative presuppositions: learning from atypical uses

The routes surveyed so far all failed to discover presupposed content. Direct tracking is unexecutable and vastly overgenerates, even if its limited record-based variant holds promise for the anaphoric subclass. Exclusion and competition at best signal that some presupposition exists, but provide nothing toward identifying it. Classic information sources seem to be completely inert. Presuppositions genuinely look undiscoverable. If such meanings are indeed learned, then some assumption or other must be revised.

We propose to abandon the assumption that presuppositional meanings are discovered from their typical uses. The learner may instead rely on atypical uses, which as we mentioned in Section 1.1 exist. Speakers sometimes use presuppositional sentences even when the presupposed content is new to their audience, relying on the audience's ability and willingness to accommodate (Stalnaker, 2002; von Stechow, 2008). Uttered to an audience that knows nothing of the speaker's pets, (10) is felicitous, and the listener learns, among other things, that the speaker has exactly two cats.

(10) Sorry I'm late! I had to take both of my cats to the vet!

Proposal. Informative uses of a trigger make discovery possible. In these cases, the presupposed content is new to the conversation, and new content can be noticed. When the content is noticeable at the moment of use, discovering its connection to the word becomes, in principle, ordinary word learning: associating a novel word with co-occurring content.

It is tempting to overestimate how useful this route is. But adults who hear (10) and learn that there are two cats do so because they already know what *both* means; they decode the content from the word. The learner, by definition, cannot. For the route to succeed, the presupposed content must be independently recoverable, from the situation (two cat carriers by the door), from the rest of the sentence, or from the surrounding discourse (an interlocutor responding, "Oh, you have *two* cats?"). Informative uses improve the odds of discovery, since they place the content where attention plausibly goes, but they do not guarantee it.

Assessment. The input must supply two things together: informative uses of the trigger, and accompanying contexts rich enough for the content to be recovered. How often the two coincide is an open empirical question, and the answer likely differs on the basis of the trigger involved, in a way that interlocks with the direct route.

Not every trigger allows informative uses (at least not to the same degree). Such uses are common for presuppositions associated with possessives, definite descriptions, change-of-state and factive verbs, etc. They are rare or even non-existent for anaphoric triggers like regular personal pronouns or discourse

particles like *too* (Kripke, 2009; Spenader, 2002; von Fintel, 2008; Roberts, 2015).⁴ This introduces a source of variability into the input that makes evidence supporting this route uneven. But there may be some structure to the variability. Triggers for which informative uses are largely unavailable are precisely those whose presuppositions may be discoverable via the direct route. Conversely, for triggers for which the direct route proves intractable, informative uses are more often available.

This route makes presuppositions noticeable, but noticeability, while a necessary condition for learning, is not sufficient. Discovery entails registering that content as part of the *word's* meaning, and a child can notice a piece of information without taking that step. Consider a child who hears (10) with the two carriers in view, and comes away knowing both that the speaker went to the vet and that she has two cats. She might still take the cat count to be superfluous information, having nothing to do with lexical meaning. Discovery, in other words, can fail after noticing succeeds, if the noticed content lands in the child's picture of the world rather than in her lexical entry for the word.

What could push noticed information into word meaning is recurrence, the observation that, across situations, a two-membered domain keeps turning up wherever *both* does. This reintroduces, in miniature, the cross-situational tracking problem of Section 3.1.1. Still, the cross-situational tracking problem is now just 'garden variety' word learning, whose difficulty can be overcome through traditional channels of information.

It is important to note that we did not solve the discovery problem as initially introduced. We chose instead to sidestep it by changing a critical assumption. Nevertheless, we think this is a promising way forward. After all, for a variety of triggers, there is simply no other clear channel of information where the content can become detectable. Notice also that this route leaves the packaging problem not just unsolved, but freshly created. Precisely because content discovered via this route was new, the child should package it as part of the assertion. Discovered this way, *both* should be assigned a conjunctive meaning as in (5b). The next section takes these conjunctive meanings as its starting point.

3.2 Packaging

If learners generally discover presuppositions via their informative uses, as we just argued, they arrive at this task with both meaning components encoded as asserted content. This is not yet target-like, but it is a far better starting position than the one discovery faced. The learner is no longer searching an open-ended space; she holds specific hypotheses and can ask a targeted question about them. Packaging remains to be solved, and where discovery had no good input in typical uses, we see plenty of ways typical uses can aid packaging. A learner can rely on content-based regularities or biases, the differential behavior of different components under embedding, and their use-conditional status at each utterance. We discuss these options below.

The packaging problem relates to an issue discussed in the formal semantics literature under 'trig-

⁴Because anaphoric triggers demand something previously made discourse-salient, their presuppositions target the history of the conversation itself. In such cases, accommodation is difficult, because the type of contextual adjustment required effectively involves a rewriting of the facts about the conversational record (von Fintel, 2008).

gering’. The triggering problem poses an ontological question about language: why are certain meaning components encoded as presuppositions rather than asserted content in the first place (Stalnaker, 1974; Abrusán, 2011; Schlenker, 2021)? We will return to the relationship with triggering in Section 4.2 and discuss packaging as a learning problem for now.

There is also a position on which the packaging problem can be completely dissolved. On certain models of presupposition (Simons, 2001; Simons et al., 2010; Tonhauser et al., 2013; Beaver et al., 2017), presuppositional behavior, such as projection through negation, is computed in-context rather than lexically stipulated. Content that is part of the word’s meaning, but happens not to be at-issue relative to the question being addressed by the utterance, projects. For the triggers this view covers, the conjunctive meaning is not an intermediate representation but the end state, and the presupposition learning problem reduces to the discovery problem. We return to this as well in Section 4.2.

3.2.1 Biases: packaging based on content

Proposal. Perhaps packaging does not need to be solved from data. It may instead be the output of a general algorithm, where content with a certain profile is routed to the presuppositional dimension by rule. Such proposals exist, though they differ on what the operative profile is: content that is not about the main point of the sentence (Abrusán, 2011), content classified as a precondition of the described event (Schlenker, 2021), or event components that fail to be cognitively privileged (e.g., for change-of-state verbs, initial states are less privileged than the changes) (Bade et al., 2024). Whatever the details, the consequence for acquisition is the same: given a discovered meaning, learners need not decide the dimension from data, because there are biases that decide it for them.

Assessment. The route’s plausibility rests on the systematicity of the phenomena, and the facts cut both ways. On one side, for some expressions, mapping tends to be stable within and across languages. Change-of-state verbs presuppose their initial states cross-linguistically (Levinson and Annamalai, 1992); cardinality restrictions on quantificational domains often surface as presuppositions (e.g., *all seven children* presupposes a domain of exactly seven). Regularities of this scope invite an algorithmic explanation. On the other side, presuppositionality is in places demonstrably arbitrary. Languages differ in which items they lexicalize. As discussed, French lacks a counterpart of *both* (Chemla, 2007); many languages lack a lexicalized counterpart of English *the*; and English, in turn, lacks the honorific presupposition triggers found in languages such as Korean, as well as the ontological-category presupposition triggers associated with rich classifier systems. An algorithm can solve the packaging problem only where it applies; for the arbitrary remainder, packaging must still be learned.

Adults productively assign presuppositions to novel material they cannot have learned item by item, including iconic gestures, which is evidence for some systematicity in packaging (Schlenker, 2021). The learning-side evidence is less favorable. Bade et al. (2024) argued from artificial word learning that learners encode novel initial states as presuppositions, which would be the bias operating during acquisition, but the replication and extension by Alstott and Aravind (2026) found no such pattern, with most

participants encoding nothing presuppositional and the remainder encoding conjunctively (Section 2.3). As matters stand, the algorithm is better supported in adult competence than in learning.

3.2.2 Projection tracking

Proposal. The second packaging route exploits the compositional signature of presupposition itself. Presuppositions project. Projection separates the target meaning from the conjunctive meaning in (5): $\langle \text{assert } a, \text{presuppose } p \rangle$ predicts that p survives embedding under entailment-canceling operators, while $a \wedge p$ does not predict this.⁵ A child who registers that “I didn’t take both cats to the vet” still implies there are two cats could draw inferences about its dimension. Likewise, recognizing that a speaker can ask “Did you take both cats to the vet?” despite already knowing that there are two cats provides evidence that the content has a different status from ordinary asserted content.

The projection route targets the packaging problem directly: it is the canonical source of evidence about presuppositionality. It is also the diagnostic that linguists themselves use. The proposal is that the child perform the same inference over naturalistic input.

Assessment. To use projection evidence, the learner must parse the embedding environment, and infer the speaker’s actual commitments in it (did the speaker of the negated sentence take the two-ness for granted, or deny it, or neither?). The main difficulty here might be data sparsity: the relevant configurations, uses of a given trigger under entailment-canceling operators in contexts where the status of the presupposed content is independently observable, may be infrequent. For *know* and *think*, this has been argued to be the case (Dudley et al., 2017). We know of no corpus estimates for other trigger classes; producing them would be a useful next step for this research program. Still, this route has the signal that is, in principle, decisive for the packaging problem.

3.2.3 Backgrounding: from conjunction to presupposition

Proposal. Suppose, again, that the learner is working with a conjunctive meaning like (5b). Backgrounding is the route by which the learner pushes one of those conjuncts to the presuppositional dimension by monitoring their discourse status. At each use, for each conjunct, the learner asks: was this content already common ground, or was it news? If one conjunct proves to be systematically established in advance while the other carries the news, the former can be relegated to the presuppositional dimension. That is, $\langle \text{assert } a \wedge p \rangle$ can be revised to $\langle \text{assert } a, \text{presuppose } p \rangle$.

Backgrounding retains the core of the direct route of Section 3.1.1, tracking the discourse status of candidate content, but is not prone to the objections raised there. The direct route asked the learner to track everything established at every use, and to search that open-ended record for diagnostic correlations. This tracking was deemed unaffordable, partially because of spurious correlates. Backgrounding demands much less. The learner arrives with the candidates in hand, the conjuncts of a meaning she

⁵Notice that a conjunctive meaning would still be *compatible* with p holding under negation: $\neg(a \wedge p)$ is true as long as one of the conjuncts is false. What would be surprising on a conjunctive meaning is systematic truth of one of the conjuncts.

already represents, and checks their status at each use. The open-ended record is replaced by a targeted query (was it already known that there are two toys?). The spurious-correlation problem does not arise, because environmental accidents were never in the candidate set to begin with; only content in the word’s hypothesized lexical entry is up for demotion. Ordering the mechanism after discovery converts an intractable search into a tractable verification.

Why should a learner, already equipped with a meaning for a word, make any changes to it? We think the driving force might be the child’s own use-conditional knowledge. As discussed in Section 2.1, children expect assertions to be informative. In the listener-choice experiments of Aravind et al. (2023), children rejected candidate addressees who already knew the asserted content of an utterance. Now consider how things look to a child who has a conjunctive meaning for *both*. On that analysis, every typical use of the word has the speaker asserting, in part, something everybody already knows. If the conjuncts are evaluated individually, the speakers around her keep violating the norm, in a systematic way: the conjunct encoding two-ness, and only that conjunct, is routinely redundant. Demoting that one conjunct to be a presupposition would make the anomaly disappear, and render those utterances unobjectionable.

We also conjecture a second, architectural reason that could drive backgrounding. The lexicon might simply disprefer maintaining meanings of the conjunctive profile (see Stalnaker, 1974 for a related idea).⁶ Even in the most classic case of complex meaning — *bachelor* defined as ‘unmarried man’ — the man component behaves like a presupposition (Thomason, 1972; for skepticism, Boër and Lycan, 1976). One can protest it using the “Hey, wait a minute” move (11), and it projects in negative contexts (12).

(11) Dr. Smith is a bachelor.

Hey, wait a minute! I didn’t know Dr. Smith was a man!

#Hey, wait a minute! I didn’t know Dr. Smith was unmarried!

(12) Dr. Smith isn’t a bachelor.

→ Dr. Smith is a man.

↗ Dr. Smith is unmarried.

If, indeed, conjunctive meanings are penalized, a learner tacitly expecting possible word meanings to respect this generalization would treat a discovered conjunctive meaning as unstable, as something to be revised, and backgrounding would supply that revision.

Assessment. Backgrounding needs three things. It needs discovered conjunctive meanings as input. It then needs *typical* uses, uses in which the presupposed content is antecedently shared, and these are, by definition, the overwhelming majority. Finally, it needs the learner to be able to assess, for a particular, already-represented piece of information, whether it was mutually known at the moment of use. This

⁶So far, we have pretended that which meanings count as conjunctive does not depend on the metalanguage of decomposition. It does: the stipulation has content only relative to a decomposition fixed on independent grounds. For instance, *kill* as ‘cause to die’ is conjunctive on some analyses, yet shows no presuppositional split. We set the issue aside here.

requirement is not trivial, but the task is of a piece with what young children demonstrably do: evaluate whether some specific content is old or new for a specific listener (Akhtar et al., 1996; Moll et al., 2007; Aravind et al., 2023).

There is no direct evidence that learners background, and one can imagine multiple ways such a process could be implemented. It might be a gradual, slow, cross-situational process, but it might also happen from very little evidence, even a single encounter.⁷ Such details aside, backgrounding, in our view, is the most general solution to the packaging problem for the bulk of the lexicon, especially if it turns out embedded uses are too sparse for projection tracking to be reliable. The demands here are modest in comparison to direct tracking, and its driving pressures are independently attested.

4 Discussion

4.1 Division of labor

Our starting point was that learning presuppositional meanings involves solving two problems. Discovery, we argued, is the hard problem. It is unsolvable if presuppositions are always old, background information. We proposed that for discovery to happen, the learner needs to encounter presuppositions used informatively, where the relevant content is new and noticeable, turning the problem into ordinary word learning (Section 3.1.4). The idea is counterintuitive: for presuppositions to be learned, they must occur often enough in contexts where they could be *misanalyzed* as assertions.

Solving discovery this way entails initially encoding the presupposed content as asserted content. To arrive at the target, the learner must also solve packaging. Part of the discovered content must be demoted to a presupposition, after the learner registers, for instance, that this part, but not the other parts, is systematically already common ground at the moment of use (Section 3.2.3). Projection evidence contributes to packaging where embedded uses occur (Section 3.2.2), and at least some content may be assigned the dimension by rule (Section 3.2.1).

The two processes neatly divide the full space of input between them: atypical informative uses, useless for backgrounding, are the evidence for discovery, and typical uses, useless for discovery, are the evidence for backgrounding. Anaphoric triggers are the limiting case in which a single record-based process can serve both components at once (Section 3.1.1), which may be why they are acquired earliest.

This view makes a few clear predictions. Discovery has to precede packaging, so learners should be observable both in a stage lacking the relevant content, and in an intermediate stage where that content is encoded assertively. The bare meanings children posit for *both* and *either*, and the conjunctive encodings in artificial learning, are candidates for these stages. Order of acquisition should track the relative availability of the different signals. For discovery, it is the rate of informative uses with recoverable content. For packaging, it is the verifiability of the discovered conjunct’s status or the amount of projection

⁷Adults’ packaging failure in the artificial learning study in Section 2.3 is not evidence against the route: the training regimes withheld precisely the signal backgrounding runs on, since the candidate content, though visible in the scene, was never established as shared between speaker and addressee before use.

evidence. Both proposed processes can be tested experimentally: training with and without informative uses should modulate discovery, and training with and without common-ground support should modulate packaging (Section 3.2.3).

4.2 Implications for linguistic and psychological theories

The discussion here has implications for theories of natural language semantics, as well as psychological theories of word learning. As we mentioned in Section 3.2, recent theoretical work on presupposition has taken up the ‘triggering problem’: the question of why certain entailments are presupposed rather than asserted. Something must distribute entailments across the two dimensions. Is it a general algorithm sensitive to the profile of the content, an item-by-item lexical stipulation, or something else? The mature system is generally a poor place to settle this question. Adult judgments display the output of the assignment, and a general algorithm and a long-fossilized stipulation generate the same outputs. It is *learning* where the assignment can be observed as it is being made.

A perspective focused on learning constrains what form a solution to triggering can take. We have argued that presupposed content can only be discovered when the presupposition does the job of assertion. The output of discovery is a conjunctive, assertive meaning. Any account on which a triggering algorithm puts a constraint on possible lexical meanings, one that bars conjunctive profiles from the hypothesis space outright, would therefore be incompatible with our solution to discovery. It would not preempt packaging so much as make discovery impossible, by forbidding the only representation through which the relevant piece of content can arrive. If we are right, then conjunctive meanings must be formable but unstable, entertained but for some reason not retained.

Importantly, the triggering problem, in its current formulation in the literature, abstracts away from what we take to be the computationally and pragmatically harder task for the learner. Discussions of triggering generally start from a word meaning whose component pieces are already in hand and ask only how they are distributed across the two dimensions. Everything prior is taken for granted. The survey above suggests that this prior problem, discovery, is where the larger difficulty in learning lives. At present, a theory of triggering solves the last, tractable step. It will be a more complete answer to why languages presuppose what they presuppose when it is coupled with an account of how presupposed content becomes representable at all.

Finally, based on what we have said of discovery (Section 3.1.4), we see a very tempting argument for a view on which presuppositional behavior is not lexically hard-wired, but derived contextually, e.g., from how a piece of content relates to the Question Under Discussion (Simons et al., 2010; Beaver et al., 2017; see also Atlas, 1977). If presupposition discovery indeed requires atypical informative uses, it seems almost a *reductio* of lexical stipulation that typical uses provide no evidence for their learning. Simons (2001) argues against lexical stipulation in a similar spirit: the presuppositions of change-of-state verbs and factives survive synonym substitution (*stop*, *quit*, and *cease* all presuppose the prior state), so they attach to the content expressed rather than to any lexical item, which favors deriving them conversationally. The contextual view resolves this tension by eliminating the difference between

atypical and typical uses. The main argument against this view comes from the very developmental evidence we summarized in Section 2.1. Children make a strong distinction between typical and atypical uses. They seem to expect the presuppositions of familiar triggers to be common ground *before* the utterance and, unlike adults, resist accommodating them (Aravind et al., 2023). It is very hard to make sense of this pattern if there is no hard-wired presupposition/assertion distinction, and presuppositions are determined in-context.

Presupposition as a class of meaning also has underappreciated consequences for psychology at large. It is a hard case for the field’s working picture of word learning and communication, due to its strong functionalist assumptions. Words are to be learned because they help communicate, aid categorization, and so forth. Such learning is supported and made useful by the same key factors: informativity, relevance, attention, the reconstruction and modulation of speakers’ intentions (Bloom, 2000). Presuppositional meaning in general sits badly in this frame. If the characteristic job of presupposition is to restate what everybody already knows, what is it even for?

Functional answers can of course be constructed. Redundancy is useful in a noisy channel, and content that must be antecedently shared gives interlocutors a way of auditing the common ground while talking about other things (Stalnaker, 2002). But it is unclear that the actual pattern of facts follows from any such story. Nothing about channel robustness predicts which contents are lexicalized as presuppositions and which are not. Nothing in it directly predicts why triggers differ in accommodability, where the anaphoric class resists the type of accommodation that other triggers routinely permit (von Stechow, 2008), or why children refuse accommodation that adults perform without effort (Aravind et al., 2023). Until a functional account can provide non-ad-hoc explanations for them, presupposition is a challenge to functionalist theories of word learning: a pervasive class of meanings without clear, predictable communicative payoff.

5 Conclusion

Presupposition triggers are words whose distinguishing content should be unlearnable. And yet the title of this paper is obviously a falsehood: children *do* learn presuppositions, sometimes by their second birthday. How? Here we proposed a solution that involved turning one problem without a solution into two, each with its own solution. The proposal remains speculative, and all evidence we have for it is about what usable correlational information might in principle exist. But it can be tested against input measurements, acquisition orderings, and the shape of intermediate representations. Many deeper questions we have left open or unaddressed. Why do natural languages so often lexicalize content to repeat what everybody already knew? And do children come to the task expecting such content, so that the routes we have described are solutions to anticipated problems? Studying development can offer new answers to old questions, and raise new questions of its own.

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